

# Variance and Central Limit Theorem for the Log Coprime Density of Random LCMs

Muhamad Wakid 

## Abstract

Let  $X_1, X_2, \dots$  be independent and identically distributed random variables uniform on  $\{2, 3, \dots, M\}$ , let  $L_k = \text{lcm}(X_1, \dots, X_k)$ , and let

$$W_k(M) := \ln(\varphi(L_k)/L_k) - \mathbb{E} \ln(\varphi(L_k)/L_k)$$

denote the centred log coprime density, with  $s_k^2 := \text{Var } W_k(M)$ . A companion paper [12] established an exact subset identity for  $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$ , a sharp asymptotic equivalence  $\rho_k(M) \rightarrow P_M$  at fixed  $M$ , and the joint-regime convergence  $\ln M \cdot (\rho_{\lfloor cM \rfloor}(M) - P_M)/P_M \rightarrow A(c)$  with  $A(c) := \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j)$ . We study second-order fluctuations in the same joint regime  $k = \lfloor cM \rfloor$ ,  $M \rightarrow \infty$ , with fixed  $c > 0$ . We prove two results. First, the variance asymptotic

$$M \ln M \cdot s_k^2 \longrightarrow B(c) := \frac{1}{e^c - 1} - \frac{1}{e^{2c} - 1},$$

via a diagonal / off-diagonal decomposition where the off-diagonal term is shown to be negligible by an exact covariance identity valid for prime pairs  $(p, q)$  with  $pq > M$ . Second, a central limit theorem:  $W_k(M)/s_k \xrightarrow{d} \mathcal{N}(0, 1)$  as  $M \rightarrow \infty$  in the joint regime. The CLT proof uses the Hall–Heyde martingale central limit theorem applied to the Doob filtration  $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$ . Two ingredients enter. A uniform increment bound  $\max_i |D_i| = O((\log M)^3/M) = o(s_k)$  on a super-polynomially likely event combines a coupon-collector tail estimate with a sieve-geometric observation  $((M/(\log M)^2)^2 > M$  for  $M$  large), either fact alone being insufficient. Conditional-variance concentration is obtained from explicit moment bounds for the hitting-time variables  $T_p := \min\{i : p \mid X_i\}$ : a per-prime variance bound  $\text{Var}(R_{pp}^{(k)}) \leq 3q_p^k/\pi_p^2$  and a cross-prime covariance bound  $|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq C_0(c)kq_p^kq_q^k$  valid when  $pq > M$  and  $\pi_p + \pi_q \leq 1/3$ , with an explicit constant  $C_0(c)$  depending only on  $c$ , together with an  $L^2$  Minkowski summation that controls the covariances among all distinct prime-pair terms.

**Keywords:** Random LCM, Euler totient, coprime density, central limit theorem, martingale central limit theorem, Doob decomposition, Hall–Heyde, probabilistic number theory.

**MSC 2020:** 60F05, 11N37, 11K65, 60G42.

## 1 Introduction

The arithmetic structure of the least common multiple of a random set of integers has been studied extensively in the regime where the support size  $n$  grows while the number of samples  $k$  stays fixed or bounded; see Cilleruelo, Rué, Šarka, and Zumalacárregui [4]; Hilberdink and Tóth [6]; Alsmeyer, Kabluchko, and Marynych [1]; Bostan, Marynych, and Raschel [2]; Buraczewski, Iksanov, and Marynych [3]; Iksanov, Marynych, and Raschel [7]; Kabluchko,

Marynych, and Raschel [8]; and Sanna [10]. These works establish limit theorems for  $\log \text{lcm}$  and for multiplicative functions of the LCM in the fixed- $k$ ,  $n \rightarrow \infty$  regime.

A complementary regime, in which the support  $\{2, \dots, M\}$  is fixed and the sample size  $k \rightarrow \infty$ , was introduced in the companion paper [12]. There the natural object of study is not the LCM itself but the coprime density

$$\frac{\varphi(L_k)}{L_k} = \prod_{p|L_k} \left(1 - \frac{1}{p}\right),$$

whose large- $k$  behaviour is governed by exactly which primes  $p \leq M$  appear as divisors of  $L_k = \text{lcm}(X_1, \dots, X_k)$ . The paper [12] established, writing  $\rho_k(M) := \mathbb{E}[\varphi(L_k)/L_k]$ ,  $P_M := \prod_{p \leq M} (1 - 1/p)$ ,  $m_p := \lfloor M/p \rfloor$ ,  $\pi_p := m_p/(M-1)$ ,  $q_p := 1 - \pi_p$ :

- *Exact identity:*

$$\rho_k(M) = P_M \sum_{T \subseteq \mathcal{P}_M} \beta_T(M)^k \prod_{p \in T} (p-1)^{-1},$$

where  $\beta_T(M) = \mathbb{P}(\gcd(X, \prod_{p \in T} p) = 1)$  for a single uniform draw  $X$ .

- *Fixed- $M$  limit with sharp rate:*  $\rho_k(M) \rightarrow P_M$  as  $k \rightarrow \infty$ , with an explicit geometric rate of convergence controlled by  $(q^{**}(M)/q^*(M))^k$ .
- *Joint-regime first-order asymptotic:* in the regime  $k = \lfloor cM \rfloor$ ,  $M \rightarrow \infty$ , fixed  $c > 0$ ,

$$\ln M \cdot \frac{\rho_k(M) - P_M}{P_M} \rightarrow A(c) := \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j).$$

**Dependence on the companion paper.** The proofs below are self-contained apart from three explicitly identified inputs from [12]: the subset-density bounds collected in Lemma 3, the level-count estimate quoted in Lemma 4, and the elementary disjointness observation quoted in Proposition 5. The exact identity and first-order asymptotic for  $\rho_k(M)$  are recalled only to situate the present second-order results and are not used in the proofs of Theorems 1 and 2. The disjointness identity needed later is also rederived directly in Lemma 7.

In this paper we study the second-order behaviour of the centred log coprime density

$$W_k(M) := \ln(\varphi(L_k)/L_k) - \mathbb{E} \ln(\varphi(L_k)/L_k),$$

in the same joint regime  $k = \lfloor cM \rfloor$ ,  $M \rightarrow \infty$ , with fixed  $c > 0$ . We prove two results.

**Theorem 1** (Variance asymptotic). *For each fixed  $c > 0$ , in the joint regime  $k = \lfloor cM \rfloor$ ,  $M \rightarrow \infty$ ,*

$$M \ln M \cdot s_k^2 \rightarrow B(c) := \sum_{j \geq 1} e^{-cj} (1 - e^{-cj}) = \frac{1}{e^c - 1} - \frac{1}{e^{2c} - 1}, \quad (1)$$

where  $s_k^2 = \text{Var } W_k(M)$ .

**Theorem 2** (Central limit theorem). *For each fixed  $c > 0$ , in the joint regime  $k = \lfloor cM \rfloor$ ,  $M \rightarrow \infty$ ,*

$$\frac{W_k(M)}{s_k} \xrightarrow{d} \mathcal{N}(0, 1).$$

Consequently,

$$\sqrt{M \ln M} \cdot \frac{W_k(M)}{\sqrt{B(c)}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \sqrt{M \ln M} \cdot \frac{\varphi(L_k)/L_k - \rho_k(M)}{P_M \sqrt{B(c)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

The first consequence follows from Theorem 1 and Slutsky's theorem; the second follows from the linear-scale transfer Lemma 18 below.

## 1.1 Method and outline

Theorem 1 is proved by a direct diagonal / off-diagonal decomposition. Writing  $X_p = \mathbf{1}\{p \mid L_k\}$  so that  $\ln(\varphi(L_k)/L_k) = \sum_{p \leq M} X_p \ln(1 - 1/p)$ , the variance decomposes as

$$s_k^2 = V_1(k, M) + V_2(k, M)$$

with  $V_1$  the diagonal variance contribution and  $V_2$  the off-diagonal covariance sum. The diagonal  $V_1$  is handled by partitioning primes by level  $m_p = j$  and applying a partial-summation analogue of Mertens' second theorem that yields  $\sum_{p: m_p=j} 1/p^2 \sim 1/(M \ln M)$  for each fixed  $j$ , giving  $M \ln M \cdot V_1 \rightarrow B(c)$  by dominated convergence. The off-diagonal  $V_2$  is bounded using the exact covariance identity

$$\beta_{\{p,q\}}(M) - q_p q_q = -\frac{m_p m_q}{(M-1)^2} \quad \text{for } pq > M,$$

together with a mean-value bound. Section 3 carries this out in detail.

Theorem 2 is proved via the Hall–Heyde martingale central limit theorem [5, Thm. 3.2], applied to the Doob filtration  $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$ . Section 4 derives the martingale decomposition  $W_k = \sum_{i=1}^k D_i$  and identifies the explicit form of the increments.

The application of [5] requires verifying two conditions. Section 5 proves the conditional Lindeberg condition via a uniform bound on the increments:  $\max_i |D_i| = O((\log M)^3/M) = o(s_k)$  on a super-polynomially likely event. The bound exploits two facts simultaneously. First, a coupon-collector tail estimate: with super-polynomially high probability, every prime  $p$  with  $m_p \geq (\log M)^2$  has been sampled by time  $k/2$ . Second, a sieve-geometric observation: once contributing primes are restricted to  $p > M/(\log M)^2$ , at most one of them can divide any given  $x \leq M$ , because  $(M/(\log M)^2)^2 > M$  for  $M$  sufficiently large. Either fact alone gives only a weak bound; together they yield the required estimate.

Section 6 proves the conditional-variance concentration  $\sum_{i=1}^k \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}] / s_k^2 \xrightarrow{P} 1$ . The argument rests on two explicit moment bounds. Lemma 14 controls the per-prime variance  $\text{Var}(R_{pp}^{(k)})$  via a substitution  $Y_p = q_p^{-2(T_p \wedge k)}$  that reduces the quantity to a function of a single geometric random variable. Lemma 15 bounds the cross-prime covariance  $|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})|$  for  $pq > M$  via a comparison between the true joint law of the hitting times  $(T_p, T_q)$  and its independent product. These bounds are combined with a single-pair second-moment estimate, a level-summation argument, and Minkowski's inequality in  $L^2$ , which controls all cross-covariances among the prime-pair summands without assuming orthogonality.

Section 7 assembles the pieces for Theorem 2. Section 8 presents numerical diagnostics for both theorems.

## 2 Preliminaries

Throughout,  $M \geq 2$  is a fixed integer,  $X_1, X_2, \dots$  are i.i.d. uniform on  $\{2, \dots, M\}$ ,  $L_k = \text{lcm}(X_1, \dots, X_k)$ , and  $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$ . For a prime  $p \leq M$ , set

$$m_p := \lfloor M/p \rfloor, \quad \pi_p := \frac{m_p}{M-1}, \quad q_p := 1 - \pi_p, \quad \xi_p^{(i)} := \mathbf{1}\{p \mid X_i\} - \pi_p.$$

Then  $m_p$  is the number of multiples of  $p$  in  $\{2, \dots, M\}$  and  $\pi_p = \mathbb{P}(p \mid X_i)$  for any single draw. For  $T \subseteq \mathcal{P}_M := \{p : p \leq M\}$ , let  $A_T(M) = \{n \in \{2, \dots, M\} : \gcd(n, \prod_{p \in T} p) = 1\}$ , so  $\beta_T(M) = |A_T(M)|/(M-1)$  is the probability that a single uniform draw is coprime to every prime in  $T$ ; in

particular  $\beta_\emptyset = 1$  and  $\beta_{\{p\}} = q_p$ . Let  $T_p := \min\{i \geq 1 : p \mid X_i\}$ , which is geometric with success probability  $\pi_p$ . Let  $\mathcal{F}_i := \sigma(X_1, \dots, X_i)$  and  $\mathcal{F}_0 := \{\emptyset, \Omega\}$ . Logarithms are natural throughout, and  $c > 0$  is fixed with  $k := \lfloor cM \rfloor$  unless otherwise stated. We write  $C$  for an absolute constant and  $C(c)$  for a constant depending only on  $c$ ; these may vary from line to line.

We record three elementary facts and the basic subset-density lemma.

**Lemma 3** (Elementary subset-density bounds; [12, Lems. 2.1–2.4]). *For every  $T \subseteq \mathcal{P}_M$ :*

- (i)  $\mathbb{P}(\text{no prime in } T \text{ divides } L_k) = \beta_T(M)^k$ ;
- (ii)  $\beta_T(M) \leq q_p(M)$  for every  $p \in T$ ,  $T \neq \emptyset$ ;
- (iii) if  $|T| = r \geq 1$ , then  $\beta_T(M)^k \leq \prod_{p \in T} q_p(M)^{k/r}$  for every real  $k > 0$ .

**Lemma 4** (Rosser–Schoenfeld level envelope; [9], [12, Lem. 6.3]). *For all  $M \geq 289$  and all  $1 \leq j \leq M/17$ ,*

$$|\{p \leq M : m_p = j\}| \leq \frac{1.26(M/j)}{\ln(M/j)}, \quad \sum_{p: m_p=j} \frac{1}{p-1} \leq \frac{5.04}{\ln(M/j)}.$$

*In particular, for  $1 \leq j \leq M^{1/2}$ ,*

$$|\{p \leq M : m_p = j\}| \leq \frac{2.52 M}{j \ln M}, \quad \sum_{p: m_p=j} \frac{1}{p-1} \leq \frac{10.08}{\ln M}.$$

**Proposition 5** (Disjointness for  $pq > M$ ; [12, Rem. 5.6]). *For distinct primes  $p, q \leq M$  with  $pq > M$ , the events  $\{p \mid X\}$  and  $\{q \mid X\}$  are disjoint under a single uniform draw  $X$  on  $\{2, \dots, M\}$ ; equivalently  $\mathbb{P}(p \mid X, q \mid X) = 0$ , so  $\mathbb{P}(p \nmid X, q \nmid X) = 1 - \pi_p - \pi_q$  and*

$$\beta_{\{p,q\}}(M) - q_p(M) q_q(M) = -\pi_p \pi_q.$$

Part (iii) of Lemma 3 is the geometric-mean bound of [12, Lem. 2.4], which underlies the sharp joint-scaling result in the companion paper.

### 3 Variance asymptotic: proof of Theorem 1

Let  $k = \lfloor cM \rfloor$ . Write  $\varphi(L_k)/L_k = \prod_{p \leq M} (1 - 1/p)^{X_p}$ , where  $X_p = \mathbf{1}\{p \mid L_k\}$ , so

$$\ln(\varphi(L_k)/L_k) = \sum_{p \leq M} X_p \ln(1 - 1/p).$$

The variance of  $X_p$  is  $(1 - q_p^k)q_p^k$  and for  $p \neq q$ ,

$$\text{Cov}(X_p, X_q) = \mathbb{P}(p \mid L_k, q \mid L_k) - (1 - q_p^k)(1 - q_q^k) = \beta_{\{p,q\}}^k - q_p^k q_q^k,$$

as follows from the inclusion-exclusion identity  $\mathbb{P}(p \mid L_k, q \mid L_k) = 1 - q_p^k - q_q^k + \beta_{\{p,q\}}^k$ . Therefore

$$s_k^2 = V_1(k, M) + V_2(k, M), \tag{2}$$

where the diagonal and off-diagonal contributions are

$$\begin{aligned} V_1(k, M) &= \sum_{p \leq M} \ln(1 - 1/p)^2 q_p^k (1 - q_p^k), \\ V_2(k, M) &= \sum_{p \neq q, p, q \leq M} \ln(1 - 1/p) \ln(1 - 1/q) (\beta_{\{p,q\}}^k - q_p^k q_q^k). \end{aligned}$$

### 3.1 Diagonal contribution

We show  $M \ln M \cdot V_1(k, M) \rightarrow B(c)$ . Partition by  $m_p = j$ :

$$V_1(k, M) = \sum_{j \geq 1} (1 - j/(M-1))^k (1 - (1 - j/(M-1))^k) \sum_{p: m_p=j} \ln(1 - 1/p)^2.$$

For  $p \in (M/(j+1), M/j]$ ,  $\ln(1 - 1/p)^2 = 1/p^2 + O(1/p^3)$ , and since  $p \geq M/(j+1)$  we have  $1/p^3 = O((j+1)^3/M^3)$ . Hence

$$\sum_{p: m_p=j} \ln(1 - 1/p)^2 = \sum_{p: m_p=j} \frac{1}{p^2} + O\left(\frac{(j+1)^3}{M^3} \cdot \pi(M/j)\right) = \sum_{p: m_p=j} \frac{1}{p^2} + O\left(\frac{(j+1)^2}{M^2 \ln M}\right).$$

**Lemma 6** (Level- $j$  sum of reciprocal squares). *For each fixed  $j \geq 1$ ,*

$$M \ln M \cdot \sum_{p: m_p=j} \frac{1}{p^2} \longrightarrow 1 \quad (M \rightarrow \infty).$$

*Proof.* By partial summation and the prime number theorem in its standard quantitative form [11],  $\sum_{p \leq x} 1/p^2 = \pi(x)/x^2 + 2 \int_2^x \pi(t)/t^3 dt$ , so  $\sum_{p \in (A, B]} 1/p^2 = \pi(B)/B^2 - \pi(A)/A^2 + 2 \int_A^B \pi(t)/t^3 dt$ . With  $A = M/(j+1)$ ,  $B = M/j$ , and using  $\pi(t) = t/\ln t + O(t/(\ln t)^2)$ ,

$$\pi(B)/B^2 = \frac{1}{B \ln B} (1 + O(1/\ln B)) = \frac{j}{M \ln(M/j)} (1 + o(1)) = \frac{j}{M \ln M} (1 + o(1)),$$

and analogously  $\pi(A)/A^2 = (j+1)/(M \ln M) (1 + o(1))$ . Their difference is

$$\pi(B)/B^2 - \pi(A)/A^2 = -\frac{1}{M \ln M} (1 + o(1)).$$

The integral is

$$2 \int_A^B \frac{1}{t^2 \ln t} (1 + O(1/\ln t)) dt = \frac{2}{\ln M} \int_A^B \frac{dt}{t^2} (1 + o(1)) = \frac{2}{\ln M} \left( \frac{1}{A} - \frac{1}{B} \right) (1 + o(1)),$$

and  $1/A - 1/B = (j+1)/M - j/M = 1/M$ , so the integral contributes  $2/(M \ln M) (1 + o(1))$ . Summing the two contributions:

$$\sum_{p: m_p=j} \frac{1}{p^2} = \left[ -\frac{1}{M \ln M} + \frac{2}{M \ln M} \right] (1 + o(1)) = \frac{1}{M \ln M} (1 + o(1)),$$

for each fixed  $j \geq 1$ . □

*Dominated convergence.* For each fixed  $j$ , Lemma 6 together with  $(1 - j/(M-1))^k \rightarrow e^{-cj}$  and  $(1 - (1 - j/(M-1))^k) \rightarrow 1 - e^{-cj}$  gives

$$M \ln M \cdot (1 - j/(M-1))^k (1 - (1 - j/(M-1))^k) \sum_{p: m_p=j} \ln(1 - 1/p)^2 \longrightarrow e^{-cj} (1 - e^{-cj}).$$

A uniform envelope follows directly from the prime-count part of Lemma 4, applied to the summand  $\ln(1 - 1/p)^2$  itself rather than to its leading term  $1/p^2$ . If  $m_p = j$ , then  $p > M/(j+1)$ , so  $p-1 > M/(2(j+1))$  whenever  $M \geq 2(j+1)$ , and hence  $\ln(1 - 1/p)^2 \leq (p-1)^{-2} < 4(j+1)^{-2}$ .

$1)^2/M^2$ . Therefore, for  $1 \leq j \leq M^{1/2}$  and  $M \geq 289$  (which guarantees both  $\ln(M/j) \geq \frac{1}{2} \ln M$  and  $M \geq 2(j+1)$  in this range),

$$\sum_{p: m_p=j} \ln(1-1/p)^2 \leq |\{p: m_p=j\}| \frac{4(j+1)^2}{M^2} \leq \frac{2.52 M}{j \ln M} \cdot \frac{4(j+1)^2}{M^2} = \frac{10.08 (j+1)^2}{M j \ln M}.$$

For the exponential factor, since  $k \geq cM - 1$  and  $1 - x \leq e^{-x}$ ,

$$(1 - j/(M-1))^k \leq e^{-kj/(M-1)} \leq e^{-cj} e^{j/(M-1)} \leq 2e^{-cj} \quad (1 \leq j \leq M^{1/2}, M \geq 5).$$

Hence each level- $j$  term of  $M \ln M \cdot V_1$  is bounded by  $10.08 (j+1)^2/j \cdot 2e^{-cj}$ , which is summable in  $j$ . The tail  $j > M^{1/2}$  decays super-exponentially, since there  $(1 - j/(M-1))^k \leq e^{-c\sqrt{M}/2}$  for  $M \geq M_0(c)$  while  $\sum_{p \leq M} \ln(1-1/p)^2 \leq \sum_{n \geq 2} (n-1)^{-2} = O(1)$ . By dominated convergence,

$$\lim_{M \rightarrow \infty} M \ln M \cdot V_1(k, M) = \sum_{j \geq 1} e^{-cj} (1 - e^{-cj}) = B(c). \quad (3)$$

### 3.2 Off-diagonal contribution is negligible

We show  $M \ln M \cdot V_2(k, M) \rightarrow 0$ . The starting point is an exact identity valid in a restricted range of prime pairs.

**Lemma 7** (Covariance identity for disjoint prime levels). *Let  $p, q \in \mathcal{P}_M$  with  $p \neq q$  and  $pq > M$ . Then*

$$\beta_{\{p,q\}}(M) - q_p(M) q_q(M) = -\frac{m_p m_q}{(M-1)^2}. \quad (4)$$

*Proof.* By inclusion-exclusion,  $|A_{\{p,q\}}(M)| = M - \lfloor M/p \rfloor - \lfloor M/q \rfloor + \lfloor M/(pq) \rfloor - 1$ . When  $pq > M$ ,  $\lfloor M/(pq) \rfloor = 0$ , so  $(M-1)\beta_{\{p,q\}} = M-1 - m_p - m_q$ . Hence

$$(M-1)^2(\beta_{\{p,q\}} - q_p q_q) = (M-1)(M-1 - m_p - m_q) - (M-1 - m_p)(M-1 - m_q) = -m_p m_q. \quad \square$$

*Remark 8.* For levels  $(j_1, j_2)$  with  $(j_1+1)(j_2+1) < M$ , every prime  $p$  with  $m_p = j_1$  and  $q$  with  $m_q = j_2$  satisfies  $pq \geq M/(j_1+1) \cdot M/(j_2+1) = M^2/((j_1+1)(j_2+1)) > M$ , so (4) applies. In particular, if  $\max(j_1, j_2) \leq M^{1/4}$  then  $(j_1+1)(j_2+1) \leq (M^{1/4}+1)^2 < M$  for all  $M \geq 7$ , so the identity applies to every pair in this range.

*Main-range bound.* Let  $\mathcal{J}(M) = \{(p, q) : p \neq q, p, q \leq M, \max(m_p, m_q) \leq M^{1/4}\}$  and write

$$V_2^{\text{main}}(k, M) := \sum_{(p,q) \in \mathcal{J}(M)} \ln(1-1/p) \ln(1-1/q) (\beta_{\{p,q\}}^k - q_p^k q_q^k),$$

with  $V_2^{\text{tail}}(k, M) := V_2(k, M) - V_2^{\text{main}}(k, M)$ .

For  $(p, q) \in \mathcal{J}(M)$ , let  $j_1 = m_p$ ,  $j_2 = m_q$ . By Remark 8 and Lemma 7,  $\beta_{\{p,q\}} - q_p q_q = -j_1 j_2 / (M-1)^2$ ; in particular  $\beta_{\{p,q\}} < q_p q_q$ , so by the mean-value form  $|a^k - b^k| \leq k|a - b| \max(a, b)^{k-1}$  with  $a = \beta_{\{p,q\}}$ ,  $b = q_p q_q$ ,

$$|\beta_{\{p,q\}}^k - q_p^k q_q^k| \leq \frac{k j_1 j_2}{(M-1)^2} (q_p q_q)^{k-1}. \quad (5)$$

Since  $q_p q_q \geq 1 - (j_1 + j_2)/(M - 1) \geq 1/2$  for  $M$  large enough (because  $j_1 + j_2 \leq 2M^{1/4} = o(M)$ ), we have  $(q_p q_q)^{k-1} \leq 2(q_p q_q)^k$ . Using also  $|\ln(1 - 1/p)| \leq 1/(p - 1)$ , the contribution of any pair  $(p, q) \in \mathcal{J}(M)$  with  $m_p = j_1, m_q = j_2$  is at most

$$\frac{1}{(p - 1)(q - 1)} \cdot \frac{2k j_1 j_2}{(M - 1)^2} (q_p q_q)^k.$$

Summing over pairs at fixed levels  $(j_1, j_2)$  — noting  $(q_p q_q)^k$  depends on the pair only through  $j_1, j_2$  — and using  $S_j(M) := \sum_{p: m_p=j} 1/(p - 1)$ ,

$$\sum_{\substack{(p,q) \in \mathcal{J}(M) \\ m_p=j_1, m_q=j_2}} \frac{1}{(p - 1)(q - 1)} \leq S_{j_1}(M) S_{j_2}(M).$$

By Lemma 4,  $S_j(M) \leq 10.08/\ln M$  uniformly for  $1 \leq j \leq M^{1/2}$  and  $M \geq 289$ , hence certainly for  $j \leq M^{1/4}$ . Moreover,  $(q_p q_q)^k \leq 2e^{-c(j_1+j_2)}$  uniformly over  $j_1, j_2 \leq M^{1/4}$  once  $M \geq M_0(c)$ : indeed  $\ln(1 - x) = -x + O(x^2)$  and  $k/(M - 1) = c + O(1/M)$  give

$$k \ln(q_p q_q) = -\frac{k(j_1 + j_2)}{M - 1} + O\left(\frac{k(j_1^2 + j_2^2)}{M^2}\right) = -c(j_1 + j_2) + O(M^{-1/2}), \quad (6)$$

uniformly in this range, since  $j_1^2 + j_2^2 \leq 2M^{1/2}$ . Therefore

$$|V_2^{\text{main}}(k, M)| \leq \frac{4k}{(M - 1)^2} \cdot \frac{(10.08)^2}{(\ln M)^2} \cdot \sum_{j_1, j_2 \geq 1} j_1 j_2 e^{-c(j_1+j_2)}.$$

The double sum factorises:

$$\sum_{j_1, j_2 \geq 1} j_1 j_2 e^{-c(j_1+j_2)} = \left( \sum_{j \geq 1} j e^{-cj} \right)^2 = \frac{e^{2c}}{(e^c - 1)^4}.$$

With  $k = \lfloor cM \rfloor$  so  $k/(M - 1)^2 = O(1/M)$ ,

$$|V_2^{\text{main}}(k, M)| = O\left(\frac{1}{M (\ln M)^2}\right). \quad (7)$$

*Tail-range bound.* For  $(p, q) \notin \mathcal{J}(M)$ , at least one of  $m_p, m_q$  exceeds  $M^{1/4}$ . Say  $m_p > M^{1/4}$ ; then  $q_p^k \leq (1 - M^{1/4}/(M - 1))^{\lfloor cM \rfloor} \leq \exp(-cM^{1/4}(1 + o(1)))$ . Using Lemma 3(ii) in the form  $\beta_{\{p,q\}} \leq q_p$ ,  $\beta_{\{p,q\}}^k \leq q_p^k$ , and the trivial bound  $q_p^k q_q^k \leq q_p^k$ ,

$$|\beta_{\{p,q\}}^k - q_p^k q_q^k| \leq 2q_p^k.$$

Writing  $\Lambda(M) := \sum_{p \leq M} 1/(p - 1) = O(\ln \ln M)$  (Mertens),

$$\begin{aligned} |V_2^{\text{tail}}(k, M)| &\leq 2 \sum_{p: m_p > M^{1/4}} \frac{q_p^k}{p - 1} \sum_{q \leq M, q \neq p} \frac{1}{q - 1} + (\text{symmetric term with } p, q \text{ swapped}) \\ &\leq 4 \Lambda(M) \exp(-cM^{1/4}(1 + o(1))) \Lambda(M) \\ &= O((\ln \ln M)^2 e^{-cM^{1/4}}), \end{aligned}$$

which vanishes super-polynomially in  $M$ . In particular  $M \ln M \cdot V_2^{\text{tail}} \rightarrow 0$ .

Combining (7) and the tail bound,

$$|V_2(k, M)| = O\left(\frac{1}{M(\ln M)^2}\right), \quad (8)$$

so  $M \ln M \cdot |V_2(k, M)| = O(1/\ln M) \rightarrow 0$ . Combined with (3), this gives (1).  $\square$

*Remark 9* (Sign of  $V_2$ ). The leading contribution to  $V_2$  comes from the region  $\max(m_p, m_q) \leq M^{1/4}$ . There  $\beta_{\{p,q\}} - q_p q_q = -m_p m_q / (M-1)^2 < 0$ , so  $\beta_{\{p,q\}}^k - q_p^k q_q^k < 0$ . The weights  $\ln(1-1/p) \ln(1-1/q)$  are positive. Hence the main-region sum, and therefore  $V_2$  for  $M$  large, is strictly negative. Numerically (Section 8)  $V_2$  is observed negative at every tested  $(k, M)$ .

*Remark 10* ( $s_k \asymp (M \ln M)^{-1/2}$ ). Theorem 1 gives  $s_k^2 \sim B(c)/(M \ln M)$  with  $B(c) > 0$  for every  $c > 0$ , so in particular  $s_k \asymp (M \ln M)^{-1/2}$  in the joint regime. This scale is used throughout the CLT proof below.

## 4 Martingale decomposition

Let  $F_k := \ln(\varphi(L_k)/L_k) = \sum_{p \leq M} \ln(1-1/p) \mathbf{1}\{p \mid L_k\}$  and  $D_i := \mathbb{E}[F_k \mid \mathcal{F}_i] - \mathbb{E}[F_k \mid \mathcal{F}_{i-1}]$  for  $1 \leq i \leq k$ .

**Theorem 11** (Doob decomposition).  $(D_i, \mathcal{F}_i)_{i=1}^k$  is a martingale-difference sequence with  $\sum_{i=1}^k D_i = W_k$ , and

$$D_i = \sum_{\substack{p \leq M \\ p \nmid L_{i-1}}} \ln(1-1/p) q_p^{k-i} \zeta_p^{(i)}. \quad (9)$$

*Proof.* By linearity,  $\mathbb{E}[F_k \mid \mathcal{F}_i] = \sum_p \ln(1-1/p) \mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i]$ . Since  $p \mid L_k$  iff  $p \mid L_i$  or  $p \mid X_j$  for some  $j > i$ , and  $X_{i+1}, \dots, X_k \perp \mathcal{F}_i$ ,

$$\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i] = \mathbf{1}\{p \mid L_i\} + \mathbf{1}\{p \nmid L_i\}(1 - q_p^{k-i}).$$

On  $\{p \mid L_{i-1}\}$ , both  $\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i]$  and  $\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_{i-1}]$  equal 1, so this prime contributes nothing to  $D_i$ . On  $\{p \nmid L_{i-1}\}$ , case analysis on  $\mathbf{1}\{p \mid X_i\} \in \{0, 1\}$  gives

$$\mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_i] - \mathbb{E}[\mathbf{1}\{p \mid L_k\} \mid \mathcal{F}_{i-1}] = \begin{cases} q_p^{k-i+1}, & p \mid X_i, \\ -q_p^{k-i}(1 - q_p), & p \nmid X_i, \end{cases}$$

which equals  $q_p^{k-i} \zeta_p^{(i)}$  in either case. Summing over  $p$  gives (9); martingale-difference and telescoping properties are immediate.  $\square$

By orthogonality of martingale increments and Theorem 1,  $\sum_{i=1}^k \mathbb{E}[D_i^2] = s_k^2 \sim B(c)/(M \ln M)$ .

## 5 Conditional Lindeberg condition

Let  $i_0 := \lceil k/2 \rceil$  and define the event

$$\Omega^* := \bigcap_{\substack{p \leq M \\ m_p \geq (\log M)^2}} \{p \mid L_{i_0-1}\}.$$



**Theorem 12** (Uniform increment bound). *For every fixed  $c > 0$  and every  $K \geq 0$ ,  $\mathbb{P}((\Omega^*)^c) = o(M^{-K})$  as  $M \rightarrow \infty$ . Moreover, there exists  $M_0 = M_0(c)$  such that, for  $M \geq M_0$ ,*

$$\max_{1 \leq i \leq k} |D_i| \leq C(c) (\log M)^3 / M = o(s_k) \quad (10)$$

on  $\Omega^*$ . More precisely, for  $i < i_0$  the bound holds almost surely, while for  $i \geq i_0$  it holds uniformly over the possible value of  $X_i$  whenever the history belongs to  $\Omega^*$ .

*Proof.* Probability of  $\Omega^*$ . For a prime  $p$  with  $m_p \geq (\log M)^2$ ,

$$\mathbb{P}(p \nmid L_{i_0-1}) = q_p^{i_0-1} \leq \exp(-\pi_p(i_0-1)) \leq \exp(-(c/2)(\log M)^2(1+o(1))).$$

A union bound over at most  $\pi(M) \leq M/\ln M$  primes gives

$$\mathbb{P}((\Omega^*)^c) \leq \frac{M}{\ln M} \exp(-(c/2)(\log M)^2(1+o(1))) = o(M^{-K})$$

for every  $K$ .

Case A:  $i \geq i_0$ , on  $\Omega^*$ . Decompose  $D_i = S_i(X_i) - C_i$ , where

$$S_i(x) := \sum_{\substack{p|x \\ p \nmid L_{i-1}}} \ln(1-1/p) q_p^{k-i}, \quad C_i := \sum_{p \nmid L_{i-1}} \ln(1-1/p) q_p^{k-i} \pi_p.$$

On  $\Omega^*$ , every contributing prime satisfies  $m_p < (\log M)^2$ , hence  $p > M/(\log M)^2$ . For  $M \geq M_0(c)$  large enough that  $M > (\log M)^4$ , any two such primes  $p_1, p_2$  obey

$$p_1 p_2 > (M/(\log M)^2)^2 = M^2/(\log M)^4 > M,$$

so  $p_1$  and  $p_2$  cannot both divide any  $x \in \{2, \dots, M\}$ . Hence at most one contributing prime divides  $x$ , giving

$$|S_i(x)| \leq \max_{\substack{p \nmid L_{i-1} \\ m_p < (\log M)^2}} \frac{q_p^{k-i}}{p-1} \leq \frac{(\log M)^2}{M-1}.$$

For  $C_i$ , a prime at level  $j \leq (\log M)^2$  contributes at most  $|\ln(1-1/p)| \pi_p \leq 1/(p-1) \cdot j/(M-1) \leq 2(j+1)/M \cdot (j+1)/M \leq 2(j+1)^2/M^2$  (using  $q_p^{k-i} \leq 1$ , the crudest factor since  $k-i$  may be as small as zero when  $i=k$ ). Summing using  $|\{p : m_p = j\}| \leq 1.26 M/(j \ln(M/j))$  of Lemma 4:

$$|C_i| \leq \sum_{j=1}^{(\log M)^2} \frac{1.26 M}{j \ln(M/j)} \cdot \frac{2(j+1)^2}{M^2} \leq \frac{5.04}{M \ln M} \sum_{j=1}^{(\log M)^2} \frac{(j+1)^2}{j} = O\left(\frac{(\log M)^3}{M}\right),$$

since  $\sum_{j=1}^J (j+1)^2/j = \sum_j (j+2+1/j) = O(J^2)$  at  $J = (\log M)^2$ .

Combining,  $|D_i| \leq |S_i(X_i)| + |C_i| \leq C(c)(\log M)^3/M$  uniformly over  $X_i$  on  $\Omega^*$ .

Case B:  $i < i_0$ , unconditionally. Put  $r := k-i$ . Since  $i < \lceil k/2 \rceil$ , one has  $r > k/2$ ; hence, for all sufficiently large  $M$ ,  $r \geq cM/3$ . We again write  $D_i = S_i(X_i) - C_i$ .

For the part of  $S_i(x)$  coming from primes  $p \geq \sqrt{M}$ , at most one such prime can divide a fixed  $x \leq M$ : two distinct prime divisors in this range would have product  $> M$ . If that prime has level  $m_p = j \leq \sqrt{M}$ , then

$$\frac{q_p^r}{p-1} \leq \frac{2(j+1)}{M} \exp\left(-\frac{rj}{M-1}\right) \leq \frac{2(j+1)}{M} e^{-cj/4} = O_c(M^{-1}),$$

uniformly in  $j$ . For primes  $p < \sqrt{M}$ , one has  $m_p \geq \sqrt{M} - 1$ , so

$$q_p^r \leq \exp\left(-\frac{rm_p}{M-1}\right) \leq e^{-c\sqrt{M}/4}$$

for large  $M$ . Consequently their total contribution is at most  $e^{-c\sqrt{M}/4} \sum_{p \leq M} (p-1)^{-1} = O(e^{-c\sqrt{M}/4} \log \log M)$ . Thus  $\sup_x |S_i(x)| = O_c(M^{-1})$ .

For the centring term, split by levels. When  $j \leq \sqrt{M}$ , Lemma 4,  $q_p^r \leq e^{-cj/4}$ ,  $\pi_p = j/(M-1)$ , and  $(p-1)^{-1} \leq 2(j+1)/M$  give

$$\sum_{p:m_p=j} |\ln(1-1/p)| \pi_p q_p^r \leq \frac{C}{M \ln M} (j+1) e^{-cj/4}.$$

Summing over  $j$  yields  $O_c((M \ln M)^{-1})$ . The levels  $j > \sqrt{M}$  contribute  $O(e^{-c\sqrt{M}/4} \log \log M)$ . Therefore  $|C_i| = O_c((M \ln M)^{-1})$  and  $|D_i| = O_c(M^{-1})$  uniformly over all histories and all values of  $X_i$ .

Comparison to  $s_k$ .  $(\log M)^3/M \cdot (M \ln M)^{1/2} = (\log M)^{7/2}/\sqrt{M} \rightarrow 0$ , establishing (10).  $\square$

**Corollary 13** (Conditional Lindeberg). *For every  $\varepsilon > 0$ ,*

$$\frac{1}{s_k^2} \sum_{i=1}^k \mathbb{E}[D_i^2 \mathbf{1}\{|D_i| > \varepsilon s_k\} \mid \mathcal{F}_{i-1}] \xrightarrow{P} 0.$$

*Proof.* By Remark 10,  $s_k \asymp (M \ln M)^{-1/2}$ , and hence  $(\log M)^3/M = o(s_k)$ . For  $M$  sufficiently large, Theorem 12 implies that every summand in the conditional Lindeberg expression vanishes on  $\Omega^*$ : for  $i < i_0$  this holds unconditionally, and for  $i \geq i_0$  it holds uniformly over  $X_i$  conditional on any history in  $\Omega^*$ . Therefore the entire nonnegative conditional Lindeberg sum is zero on  $\Omega^*$ . Consequently, for every  $\delta > 0$ ,

$$\mathbb{P}\left(\frac{1}{s_k^2} \sum_{i=1}^k \mathbb{E}[D_i^2 \mathbf{1}\{|D_i| > \varepsilon s_k\} \mid \mathcal{F}_{i-1}] > \delta\right) \leq \mathbb{P}((\Omega^*)^c) \rightarrow 0.$$

$\square$

## 6 Conditional-variance concentration

Let  $U_i := \mathbb{E}[D_i^2 \mid \mathcal{F}_{i-1}]$ . Expanding via (9),

$$\sum_{i=1}^k U_i = A + 2B$$

where

$$A := \sum_{p \leq M} c_{pp} R_{pp}^{(k)}, \quad B := \sum_{p < q \leq M} c_{pq} R_{pq}^{(k)},$$

with  $c_{pq} := \ln(1-1/p) \ln(1-1/q) \kappa_{pq}$ ,  $\kappa_{pp} := \pi_p q_p$ ,  $\kappa_{pq} := \beta_{\{p,q\}} - q_p q_q$  for  $p \neq q$ , and

$$R_{pp}^{(k)} := \sum_{i=1}^k \mathbf{1}\{T_p \geq i\} q_p^{2(k-i)}, \quad R_{pq}^{(k)} := \sum_{i=1}^k \mathbf{1}\{T_p \geq i, T_q \geq i\} (q_p q_q)^{k-i}.$$

By Proposition 5,  $\kappa_{pq} = -\pi_p \pi_q$  whenever  $pq > M$ .

The main estimates of this section are the following.

**Lemma 14** (Per-prime variance). *For every  $M \geq 51$ , every prime  $p \leq M$ , and every  $k \geq 1$ ,*

$$\text{Var}(R_{pp}^{(k)}) \leq \frac{3q_p^k}{\pi_p^2}. \quad (11)$$

**Lemma 15** (Cross-prime covariance). *There exists a constant  $C_0(c) > 0$  (depending only on  $c$ ) such that, for distinct primes  $p, q \leq M$  with  $pq > M$  and  $\pi_p + \pi_q \leq 1/3$ , in the joint regime  $k = \lfloor cM \rfloor$ ,*

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq C_0(c) k q_p^k q_q^k. \quad (12)$$

An admissible value is  $C_0(c) = 57 + 54/c$ ; this is what the proof chain yields directly, though numerical evidence suggests that it is not sharp. Only the shape of the bound — linear in  $k$ , multiplicative in  $q_p^k q_q^k$  — is used in Theorem 17 below.

## 6.1 Proof of Lemma 14

Set  $Y_p := q_p^{-2(T_p \wedge k)}$ . Then  $R_{pp}^{(k)} = q_p^{2k}(Y_p - 1)/(1 - q_p^2)$ , and

$$\text{Var}(R_{pp}^{(k)}) = \frac{q_p^{4k} \text{Var}(Y_p)}{(1 - q_p^2)^2} \leq \frac{q_p^{4k} \mathbb{E}[Y_p^2]}{(1 - q_p^2)^2}.$$

Compute

$$\mathbb{E}[Y_p^2] = \sum_{m=1}^k q_p^{-4m} \pi_p q_p^{m-1} + q_p^{-3k} = \frac{\pi_p}{q_p} \cdot \frac{q_p^{-3k} - 1}{1 - q_p^3} + q_p^{-3k}.$$

Since  $1 - q_p^3 = \pi_p(1 + q_p + q_p^2)$ , the first term simplifies to  $(q_p^{-3k} - 1)/(q_p(1 + q_p + q_p^2)) \leq q_p^{-3k}/(q_p(1 + q_p + q_p^2))$ . Every prime  $p \geq 2$  has  $m_p \leq M/2$ , so  $\pi_p \leq (M/2)/(M-1) \leq 51/100$  for  $M \geq 51$ , and hence  $q_p \geq 49/100$  for every prime  $p \leq M$ . Consequently  $q_p(1 + q_p + q_p^2) \geq 0.49(1 + 0.49 + 0.49^2) > 4/5$ , so

$$\mathbb{E}[Y_p^2] \leq q_p^{-3k}(5/4 + 1) = (9/4) q_p^{-3k} \leq 3 q_p^{-3k}.$$

Using  $(1 - q_p^2)^2 = \pi_p^2(1 + q_p)^2 \geq \pi_p^2$ :

$$\text{Var}(R_{pp}^{(k)}) \leq \frac{q_p^{4k} \cdot 3q_p^{-3k}}{\pi_p^2} = \frac{3q_p^k}{\pi_p^2}.$$

## 6.2 Proof of Lemma 15

Write  $\mathbb{P}_{\text{dep}}$  for the joint law of  $(T_p, T_q)$  under the true distribution and  $\mathbb{P}_{\text{ind}}$  for the product of the (common) marginals. By Proposition 5, at each step three outcomes are possible with probabilities  $\beta, \pi_p, \pi_q$  (where  $\beta := \beta_{\{p,q\}} = 1 - \pi_p - \pi_q$ ), giving, for  $1 \leq i < j \leq k$ ,

$$\mathbb{P}_{\text{dep}}(T_p = i, T_q = j) = \beta^{i-1} \pi_p q_p^{j-i-1} \pi_q,$$

and the analogous formula for  $j < i$ , with  $\mathbb{P}_{\text{dep}}(T_p = T_q < \infty) = 0$  (disjointness). Under  $\mathbb{P}_{\text{dep}}$ , the marginal law of  $T_p$  is geometric with parameter  $\pi_p$  (at each step the event  $\{p \mid X\}$  has probability  $\pi_p$ ), and similarly for  $T_q$ . The product law with identical marginals is then

$$\mathbb{P}_{\text{ind}}(T_p = i, T_q = j) = \pi_p q_p^{i-1} \cdot \pi_q q_q^{j-1} = \pi_p \pi_q (q_p q_q)^{i-1} q_q^{j-i} \quad (i < j),$$

and analogously for  $j < i$  and for the diagonal  $\mathbb{P}_{\text{ind}}(T_p = T_q = m) = \pi_p \pi_q (q_p q_q)^{m-1}$ . Set  $\delta := \pi_p \pi_q / (q_p q_q)$ , so  $\beta / (q_p q_q) = 1 - \delta$ . Under  $\pi_p + \pi_q \leq 1/3$ ,  $q_p q_q \geq (1 - \pi_p)(1 - \pi_q) \geq 1 - (\pi_p + \pi_q) \geq 2/3$ , whence  $\delta \leq (3/2)\pi_p \pi_q$  and  $1/q_p, 1/q_q \leq 3/2$ . Since marginals agree,  $\text{Cov}_{\text{dep}}(Y_p, Y_q) = \mathbb{E}_{\text{dep}}[Y_p Y_q] - \mathbb{E}_{\text{ind}}[Y_p Y_q]$ , where  $Y_p := q_p^{-2(T_p \wedge k)}$ ,  $Y_q := q_q^{-2(T_q \wedge k)}$ .

Partition  $\{(i, j) : 1 \leq i, j \leq k+1\}$  (with  $T > k$  encoded as  $T = k+1$ ) into the regions  $i < j$ ,  $i > j$ ,  $i = j \leq k$ , and  $i = j = k+1$ .

Region  $i < j$ . The density ratio is

$$\frac{\mathbb{P}_{\text{dep}}(T_p = i, T_q = j)}{\mathbb{P}_{\text{ind}}(T_p = i, T_q = j)} = \frac{(1 - \delta)^{i-1}}{q_q}.$$

Write  $(1 - \delta)^{i-1}/q_q - 1 = [(1 - \delta)^{i-1} - 1]/q_q + \pi_q/q_q$ . Using  $|(1 - \delta)^{i-1} - 1| \leq (i - 1)\delta$ ,

$$\begin{aligned} & |\mathbb{E}_{\text{dep}}[Y_p Y_q; i < j] - \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j]| \\ & \leq (\delta/q_q) \mathbb{E}_{\text{ind}}[(T_p - 1)Y_p Y_q; i < j] + (\pi_q/q_q) \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j]. \end{aligned}$$

Under  $\mathbb{P}_{\text{ind}}$ , independence gives  $\mathbb{E}[T_p Y_p Y_q; i < j] \leq \mathbb{E}[T_p Y_p] \mathbb{E}[Y_q]$ . For  $r = q_p^{-1} > 1$ :  $\sum_{m=1}^k m r^m \leq k r^{k+1} / (r - 1)$  (telescoping identity  $(r - 1) \sum_m m r^m = k r^{k+1} - \sum_m r^m$ , dropping the positive term). Hence

$$\mathbb{E}[T_p Y_p] = \frac{\pi_p}{q_p} \sum_{m=1}^k m q_p^{-m} + (k+1) q_p^{-k} \leq \frac{\pi_p}{q_p} \cdot \frac{k q_p^{-k-1}}{\pi_p} + 2k q_p^{-k} = k q_p^{-k-1} + 2k q_p^{-k} \leq 4k q_p^{-k},$$

for  $q_p \geq 1/2$ . Analogously  $\mathbb{E}[Y_q] \leq 3q_q^{-k}$  (from  $\mathbb{E}[Y_q] = (1 + 1/q_q)q_q^{-k} - 1/q_q \leq 3q_q^{-k}$ ) and  $\mathbb{E}[Y_p] \leq 3q_p^{-k}$ . Thus

$$\mathbb{E}_{\text{ind}}[T_p Y_p Y_q; i < j] \leq 12k q_p^{-k} q_q^{-k}, \quad \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j] \leq 9 q_p^{-k} q_q^{-k}.$$

Using  $\delta/q_q \leq (3/2)(3/2)\pi_p \pi_q = (9/4)\pi_p \pi_q$  and  $\pi_q/q_q \leq (3/2)\pi_q$ :

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; i < j] - \mathbb{E}_{\text{ind}}[Y_p Y_q; i < j]| \leq 27k \pi_p \pi_q q_p^{-k} q_q^{-k} + \frac{27}{2} \pi_q q_p^{-k} q_q^{-k}. \quad (13)$$

Region  $i > j$ . By symmetry (swap roles of  $p$  and  $q$ ):

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; i > j] - \mathbb{E}_{\text{ind}}[Y_p Y_q; i > j]| \leq 27k \pi_p \pi_q q_p^{-k} q_q^{-k} + \frac{27}{2} \pi_p q_p^{-k} q_q^{-k}. \quad (14)$$

Region  $i = j \leq k$ . Here  $\mathbb{P}_{\text{dep}} = 0$  and  $\mathbb{P}_{\text{ind}}(T_p = T_q = m) = \pi_p \pi_q (q_p q_q)^{m-1}$ , contributing

$$\mathbb{E}_{\text{ind}}[Y_p Y_q; T_p = T_q \leq k] = \pi_p \pi_q \sum_{m=1}^k (q_p q_q)^{m-1} (q_p q_q)^{-2m} = \pi_p \pi_q (q_p q_q)^{-1} \sum_{m=1}^k (q_p q_q)^{-m},$$

and using  $\sum_{m=1}^k r^m \leq k r^k$  for  $r = (q_p q_q)^{-1} \geq 1$ ,

$$\mathbb{E}_{\text{ind}}[Y_p Y_q; T_p = T_q \leq k] \leq \pi_p \pi_q k (q_p q_q)^{-k-1} \leq \frac{3}{2} k \pi_p \pi_q q_p^{-k} q_q^{-k}. \quad (15)$$

Region  $i = j = k+1$ . Here  $\mathbb{P}_{\text{dep}}(T_p > k, T_q > k) = \beta^k$  and  $\mathbb{P}_{\text{ind}}(T_p > k, T_q > k) = (q_p q_q)^k$ , with  $Y_p Y_q = q_p^{-2k} q_q^{-2k}$ :

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; \text{both} > k] - \mathbb{E}_{\text{ind}}[Y_p Y_q; \text{both} > k]| = (q_p q_q)^{-2k} |\beta^k - (q_p q_q)^k|.$$

Using  $|\beta^k - (q_p q_q)^k| = (q_p q_q)^k |1 - (1 - \delta)^k| \leq (q_p q_q)^k k \delta$  and  $\delta \leq (3/2)\pi_p \pi_q$ ,

$$|\mathbb{E}_{\text{dep}}[Y_p Y_q; \text{both} > k] - \mathbb{E}_{\text{ind}}[Y_p Y_q; \text{both} > k]| \leq \frac{3}{2} k \pi_p \pi_q (q_p q_q)^{-k} \leq \frac{3}{2} k \pi_p \pi_q q_p^{-k} q_q^{-k}. \quad (16)$$

Assembly. Combining (13)–(16),

$$|\text{Cov}(Y_p, Y_q)| \leq 57 k \pi_p \pi_q q_p^{-k} q_q^{-k} + \frac{27}{2} (\pi_p + \pi_q) q_p^{-k} q_q^{-k}. \quad (17)$$

(The coefficient  $57 = 27 + 27 + 3/2 + 3/2$  dominates the individual contributions.) Using the linear relation  $R_{pp}^{(k)} = q_p^{2k} (Y_p - 1) / (1 - q_p^2)$  from the proof of Lemma 14 (analogously for  $q$ ) and  $(1 - q_p^2)(1 - q_q^2) = \pi_p \pi_q (1 + q_p)(1 + q_q) \geq \pi_p \pi_q$ :

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| = \frac{(q_p q_q)^{2k} |\text{Cov}(Y_p, Y_q)|}{(1 - q_p^2)(1 - q_q^2)} \leq \frac{|\text{Cov}(Y_p, Y_q)| (q_p q_q)^{2k}}{\pi_p \pi_q}.$$

Substituting (17):

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq \left[ 57k + \frac{27}{2} \left( \frac{1}{\pi_q} + \frac{1}{\pi_p} \right) \right] q_p^k q_q^k.$$

Since  $\pi_p \geq 1/(M-1)$  trivially and  $k = \lfloor cM \rfloor \geq cM - 1$ , we have  $M \leq (k+1)/c$ , so  $1/\pi_p \leq M-1 \leq (k+1)/c$ , and likewise for  $q$ . Therefore  $1/\pi_p + 1/\pi_q \leq 2(k+1)/c$ , and using  $k+1 \leq 2k$  for  $k \geq 1$ ,

$$|\text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq [57 + 54/c] k q_p^k q_q^k.$$

Thus  $C_0(c) := 57 + 54/c$  suffices for all  $k \geq 1$ .  $\square$

### 6.3 Level-summation bound and assembly

**Lemma 16** (Level-summation upper bound). *Let  $f: \mathbb{N}_{\geq 1} \rightarrow \mathbb{R}_{\geq 0}$  satisfy  $f(j) \leq A(1+j)^r$  for  $A \geq 0$ ,  $r \geq 0$ . For each  $c > 0$  there exist  $M_0(c, r)$  and  $C(c, r)$  such that for  $M \geq M_0(c, r)$ ,*

$$L_M(f) := \sum_{p \leq M} \frac{f(m_p) q_p^{\lfloor cM \rfloor}}{(p-1)^4} \leq \frac{C(c, r) A}{M^3 \ln M}. \quad (18)$$

*Proof.* Partitioning by level and bounding  $1/(p-1)^4 \leq 16(j+1)^4/M^4$  for  $M \geq 2(j+1)$ , together with Lemma 4:

$$S_j(M) := \sum_{p: m_p=j} \frac{1}{(p-1)^4} \leq \frac{20.16(j+1)^4}{j \ln(M/j) M^3}.$$

For  $j \leq M^{1/2}$ ,  $\ln(M/j) \geq (\ln M)/2$ , so  $S_j(M) \leq 40.32(j+1)^4/(j \ln M \cdot M^3)$ .

For the exponential factor,  $\ln(1-x) \leq -x$  (concavity) gives  $\ln(1-j/(M-1))^{\lfloor cM \rfloor} \leq -cj + O(j/M)$  uniformly for  $j \leq M^{1/2}$ , so  $(1-j/(M-1))^{\lfloor cM \rfloor} \leq 2e^{-cj}$  for  $M \geq M_1(c)$ .

Main range  $j \leq M^{1/2}$ :

$$\sum_{j=1}^{\lfloor M^{1/2} \rfloor} f(j) \cdot (1-j/(M-1))^{\lfloor cM \rfloor} S_j(M) \leq \frac{80.64}{M^3 \ln M} \sum_{j \geq 1} \frac{A(1+j)^{r+4} e^{-cj}}{j},$$

and the series converges to a finite  $\Sigma(c, r) < \infty$ .

Tail  $j > M^{1/2}$ :  $q_p^k \leq e^{-c\sqrt{M}/2}$  times  $\sum_p 1/(p-1)^4 = O(1)$  times  $\sum_{j > M^{1/2}}^M A(1+j)^r = O(AM^{r+1})$ , giving super-polynomially small total. Absorbing into the main-range bound yields (18) with  $C(c, r) = 81 \Sigma(c, r)$ .  $\square$

**Theorem 17** (Conditional-variance concentration). *As  $M \rightarrow \infty$  in the joint regime,*

$$\text{Var}\left(\sum_{i=1}^k U_i\right) = O_c\left(\frac{1}{M^2(\ln M)^4}\right) = o(s_k^4).$$

Consequently  $\sum_i U_i / s_k^2 \xrightarrow{P} 1$ .

*Proof.* We have

$$\text{Var}\left(\sum_i U_i\right) = \text{Var}(A) + 4 \text{Cov}(A, B) + 4 \text{Var}(B).$$

Set  $J := M^{1/4}$ . Terms involving a prime with  $m_p > J$  will be called tail terms. Since

$$q_p^k \leq \exp\left(-\frac{km_p}{M-1}\right) \leq e^{-cJ/2}$$

for such primes and all sufficiently large  $M$ , every tail estimate below is super-polynomially small; polynomial factors in  $M$  and powers of  $\log \log M$  are therefore harmless.

*Bound on  $\text{Var}(A)$ .* By Lemma 14, which applies to every prime  $p \leq M$  once  $M \geq 51$ ,

$$c_{pp}^2 \text{Var}(R_{pp}^{(k)}) \leq \frac{\pi_p^2}{(p-1)^4} \frac{3q_p^k}{\pi_p^2} = \frac{3q_p^k}{(p-1)^4}.$$

Lemma 16 therefore gives

$$\sum_p c_{pp}^2 \text{Var}(R_{pp}^{(k)}) = O_c\left(\frac{1}{M^3 \ln M}\right).$$

For the covariance terms with  $m_p, m_q \leq J$ , we have  $pq > M$  and  $\pi_p + \pi_q \leq 1/3$  for all sufficiently large  $M$ , so Lemma 15 applies. Hence

$$\sum_{\substack{p \neq q \\ m_p, m_q \leq J}} |c_{pp} c_{qq} \text{Cov}(R_{pp}^{(k)}, R_{qq}^{(k)})| \leq C_0(c)k \left( \sum_p \frac{\pi_p q_p^k}{(p-1)^2} \right)^2 = O_c\left(\frac{1}{M^3 (\ln M)^2}\right),$$

To justify the last estimate explicitly, set

$$S_M := \sum_{p \leq M} \frac{\pi_p q_p^k}{(p-1)^2}.$$

At level  $m_p = j \leq M^{1/2}$ , use  $\pi_p = j/(M-1)$ ,  $(p-1)^{-2} \leq 4(j+1)^2/M^2$ , the count in Lemma 4, and  $q_p^k \leq 2e^{-cj}$  to obtain

$$S_M \leq \frac{C}{M^2 \ln M} \sum_{j \geq 1} (j+1)^2 e^{-cj} + O(e^{-c\sqrt{M}/2}) = O_c\left(\frac{1}{M^2 \ln M}\right).$$

Since  $k = O_c(M)$ , this gives  $kS_M^2 = O_c(M^{-3}(\ln M)^{-2})$  as claimed. If at least one of  $m_p, m_q$  exceeds  $J$ , Cauchy–Schwarz and Lemma 14 give

$$|c_{pp}| \sqrt{\text{Var}(R_{pp}^{(k)})} \leq \frac{\sqrt{3} q_p^{k/2}}{(p-1)^2},$$

and the resulting double sum is  $O_c(e^{-cJ/4})$ . Thus

$$\text{Var}(A) = O_c\left(\frac{1}{M^3 \ln M}\right). \quad (19)$$

*A single-pair bound for  $R_{pq}^{(k)}$ .* First suppose  $m_p, m_q \leq J$ . Then  $pq > M$ , so the events  $\{p \mid X\}$  and  $\{q \mid X\}$  are disjoint. Let  $\beta = 1 - \pi_p - \pi_q$ ,  $a = q_p q_q$ , and  $N = (T_p \wedge T_q) \wedge k$ . Since

$$R_{pq}^{(k)} = \frac{a^k(a^{-N} - 1)}{1 - a},$$

we can bound its second moment directly. Put  $u := \pi_p + \pi_q$  and  $d := \pi_p \pi_q$ , so that  $\beta = 1 - u$  and  $a = 1 - u + d$ . Because  $u \leq 1/3$  and  $d \leq u^2/4$ , one has  $a^2 \leq \beta$  and  $1 - a = u - d \geq u/2$ . Therefore

$$\begin{aligned} a^{2k} \mathbb{E}[a^{-2N}] &= u \sum_{m=1}^k a^{2(k-m)} \beta^{m-1} + \beta^k \\ &\leq ku\beta^{k-1} + \beta^k \leq C(c)ku\beta^{k-1}, \end{aligned}$$

where the last step uses  $ku \geq c/2$  for all sufficiently large  $M$ . Since  $(a^{-N} - 1)^2 \leq a^{-2N}$ ,

$$\text{Var}(R_{pq}^{(k)}) \leq \mathbb{E}[(R_{pq}^{(k)})^2] \leq \frac{C(c)k\beta^{k-1}}{\pi_p + \pi_q}. \quad (20)$$

At a level pair  $(j_1, j_2)$  with  $j_1, j_2 \leq J$  one has  $\pi_p + \pi_q = (j_1 + j_2)/(M - 1)$ , and, since  $1 - x \leq e^{-x}$  and  $(k - 1)/(M - 1) = c + O(1/M)$ ,

$$\beta^{k-1} \leq \exp\left(-\frac{(k-1)(j_1 + j_2)}{M-1}\right) \leq 2e^{-c(j_1 + j_2)}$$

for  $M \geq M_0(c)$ , uniformly over  $j_1, j_2 \leq J$  (as in (6)). Together with  $|c_{pq}| \leq \pi_p \pi_q / ((p-1)(q-1)) \leq 16(1+j_1)^2(1+j_2)^2/M^4$ , the bound (20) gives

$$|c_{pq}| \sqrt{\text{Var}(R_{pq}^{(k)})} \leq \frac{C(c)}{M^3} \frac{(1+j_1)^2(1+j_2)^2}{\sqrt{j_1 + j_2}} e^{-c(j_1 + j_2)/2}. \quad (21)$$

*Bound on  $\text{Var}(B)$ .* Write  $B = B_{\text{main}} + B_{\text{tail}}$ , where  $B_{\text{main}}$  contains pairs with  $m_p, m_q \leq J$ . Minkowski's inequality in  $L^2$  gives

$$\sqrt{\text{Var}(B_{\text{main}})} \leq \sum_{\substack{p < q \\ m_p, m_q \leq J}} |c_{pq}| \sqrt{\text{Var}(R_{pq}^{(k)})}.$$

By Lemma 4, the number of primes at level  $j \leq J$  is at most  $2.52M/(j \ln M)$ . Combining this count with (21),

$$\sqrt{\text{Var}(B_{\text{main}})} \leq \frac{C(c)}{M(\ln M)^2} \sum_{j_1, j_2 \geq 1} \frac{(1+j_1)^2(1+j_2)^2}{j_1 j_2 \sqrt{j_1 + j_2}} e^{-c(j_1 + j_2)/2} = O_c\left(\frac{1}{M(\ln M)^2}\right),$$

because the double series converges.

It remains to control  $B_{\text{tail}}$ , which also includes every pair with  $pq \leq M$ . For arbitrary distinct  $p, q$ , let

$$\beta_{pq} := \mathbb{P}(p \nmid X, q \nmid X), \quad a := q_p q_q, \quad N := (T_p \wedge T_q) \wedge k.$$

If  $p$  is chosen so that  $m_p = \max(m_p, m_q)$ , then  $a \leq q_p$  and  $\beta_{pq} \leq q_p$ . Directly from the geometric law of  $T_p \wedge T_q$ ,

$$a^{2k} \mathbb{E}[a^{-2N}] = (1 - \beta_{pq}) \sum_{m=1}^k a^{2(k-m)} \beta_{pq}^{m-1} + \beta_{pq}^k \leq (k+1)q_p^{k-1}.$$

Since  $1 - a \geq \pi_p$ ,

$$\text{Var}(R_{pq}^{(k)}) \leq \mathbb{E}[(R_{pq}^{(k)})^2] \leq \frac{(k+1)q_p^{k-1}}{\pi_p^2}. \quad (22)$$

For every tail pair,  $m_p > J$  after relabelling. Using the crude bounds  $|\kappa_{pq}| \leq 1$ ,  $|\ln(1 - 1/r)| \leq 1/(r-1)$ ,  $1/\pi_p \leq M$ , and  $\sum_{\substack{r \leq M \\ r \text{ prime}}} (r-1)^{-1} = O(\log \log M)$ , Minkowski's inequality and (22) give

$$\sqrt{\text{Var}(B_{\text{tail}})} \leq CM^{3/2}(\log \log M)^2 e^{-cJ/4} = o(M^{-K})$$

for every fixed  $K$ . Therefore

$$\text{Var}(B) = O_c\left(\frac{1}{M^2(\ln M)^4}\right). \quad (23)$$

This argument controls all covariances among distinct prime-pair summands through Minkowski's inequality; no unproved orthogonality is used.

*Conclusion.* By Cauchy-Schwarz and (19)–(23),

$$|\text{Cov}(A, B)| \leq \sqrt{\text{Var}(A) \text{Var}(B)} = O_c\left(\frac{1}{M^{5/2}(\ln M)^{5/2}}\right) = o\left(\frac{1}{M^2(\ln M)^2}\right).$$

Thus

$$\text{Var}\left(\sum_i U_i\right) = O_c\left(\frac{1}{M^2(\ln M)^4}\right) = o(s_k^4),$$

because  $s_k^4 \asymp M^{-2}(\ln M)^{-2}$  by Theorem 1. Finally,  $\mathbb{E}[\sum_i U_i] = \sum_i \mathbb{E}[D_i^2] = s_k^2$ , so Chebyshev's inequality gives  $\sum_i U_i / s_k^2 \xrightarrow{P} 1$ .  $\square$

**Lemma 18** (Linear-scale transfer). *Let  $Y_k := \varphi(L_k)/L_k$  and  $\mu_k := \mathbb{E} \ln Y_k$ . In the joint regime,*

$$\frac{e^{\mu_k}}{P_M} \longrightarrow 1, \quad \frac{Y_k - \rho_k(M)}{P_M s_k} - \frac{W_k(M)}{s_k} \xrightarrow{P} 0.$$

Consequently the log-scale central limit theorem implies the linear-scale central limit theorem stated in Theorem 2.

*Proof.* Writing  $a_p := \ln(1 - 1/p) < 0$ ,

$$\mu_k - \ln P_M = - \sum_{p \leq M} a_p q_p^k = \sum_{p \leq M} |a_p| q_p^k.$$

Partitioning by  $m_p = j$ , using  $|a_p| \leq 1/(p-1)$  and Lemma 4 for  $j \leq M^{1/2}$ , gives

$$\sum_{p \leq M} |a_p| q_p^k \leq \frac{C(c)}{\ln M} \sum_{j \geq 1} e^{-cj} + o(1/\ln M) = O_c(1/\ln M).$$



The range  $j > M^{1/2}$  is super-polynomially small. Hence  $e^{\mu_k}/P_M = 1 + O_c(1/\ln M)$ . By Mertens' product theorem [11],  $P_M \asymp 1/\ln M$ , so also  $e^{-\mu_k} = O(\ln M)$ .

Since  $Y_k = e^{\mu_k + W_k} \leq 1$ , we have  $e^{W_k} \leq e^{-\mu_k}$ . For  $R_k := e^{W_k} - 1 - W_k$ , Taylor's inequality gives

$$0 \leq R_k \leq \frac{1}{2}e^{(W_k)_+}W_k^2 \leq \frac{1}{2}e^{-\mu_k}W_k^2.$$

Therefore

$$\frac{\mathbb{E}|R_k|}{s_k} \leq \frac{1}{2}e^{-\mu_k}s_k = O\left(\sqrt{\frac{\ln M}{M}}\right) \rightarrow 0,$$

where Theorem 1 was used. Since  $\rho_k(M) = e^{\mu_k}\mathbb{E}[e^{W_k}]$ ,

$$\frac{Y_k - \rho_k(M)}{P_M s_k} = \frac{e^{\mu_k}}{P_M} \left( \frac{W_k}{s_k} + \frac{R_k - \mathbb{E}R_k}{s_k} \right).$$

The second term in parentheses tends to zero in  $L^1$ , while  $W_k/s_k$  is tight because it has variance one. Together with  $e^{\mu_k}/P_M \rightarrow 1$ , this proves the claim. The same estimate also yields the Jensen-gap bound  $0 \leq \rho_k(M) - e^{\mu_k} \leq s_k^2/2 = o(P_M s_k)$ .  $\square$

## 7 Proof of Theorem 2

For each  $M$ , define the row of the triangular array by  $Z_{M,i} := D_i/s_k$ ,  $1 \leq i \leq k = \lfloor cM \rfloor$ , with row filtration  $\mathcal{F}_{M,i} := \mathcal{F}_i$ . Theorem 11 gives  $\mathbb{E}[Z_{M,i} \mid \mathcal{F}_{M,i-1}] = 0$  and  $\sum_i Z_{M,i} = W_k/s_k$ . We use the Hall–Heyde martingale central limit theorem [5, Thm. 3.2] in the following standard form: it is enough that, for every  $\varepsilon > 0$ ,

$$\sum_{i=1}^k \mathbb{E}[Z_{M,i}^2 \mathbf{1}\{|Z_{M,i}| > \varepsilon\} \mid \mathcal{F}_{M,i-1}] \xrightarrow{P} 0,$$

and that

$$\sum_{i=1}^k \mathbb{E}[Z_{M,i}^2 \mid \mathcal{F}_{M,i-1}] \xrightarrow{P} 1.$$

Corollary 13 verifies the first condition and Theorem 17 verifies the second. Hence  $W_k/s_k \xrightarrow{d} \mathcal{N}(0, 1)$ . Theorem 1 and Slutsky's theorem give  $\sqrt{M \ln M} W_k / \sqrt{B(c)} \xrightarrow{d} \mathcal{N}(0, 1)$ . Lemma 18 gives the corresponding linear-scale limit for  $\varphi(L_k)/L_k - \rho_k(M)$ .  $\square$

## 8 Numerical Evidence

The theorems in Sections 3–7 are proved rigorously and require no numerical verification. The numerics below are included to illustrate finite- $M$  behaviour, and for Theorem 2 to document that descriptive diagnostics of Gaussianity are consistent with the proof. The variance calculation evaluates the exact finite formula in extended-precision floating-point arithmetic, with an independent 80-decimal arbitrary-precision spot check. Monte Carlo sampling uses NumPy's default PRNG, freshly initialised with the fixed seed 2026 for each panel.

*Theorem 1.* The rescaled variance  $M \ln M \cdot s_k^2$  is evaluated deterministically from the exact finite formula  $s_k^2 = V_1 + V_2$  via direct summation over  $\mathcal{P}_M$  and  $\mathcal{P}_M \times \mathcal{P}_M$ . Figure 1 plots  $M \ln M \cdot s_k^2$  versus  $M$  at  $k = \lfloor cM \rfloor$  for  $c \in \{0.5, 1, 2, 3\}$  and  $M$  up to 3200. Table 1 provides representative

numerical values. Convergence to  $B(c)$  is slow and visibly nonmonotone at finite  $M$ : at  $M = 3200$ , the rescaled variance agrees with  $B(c)$  to within approximately 2% across  $c \in [0.5, 3]$ . The off-diagonal contribution  $V_2$  is always negative (consistent with Remark 9) and an order of magnitude smaller than  $V_1$ .

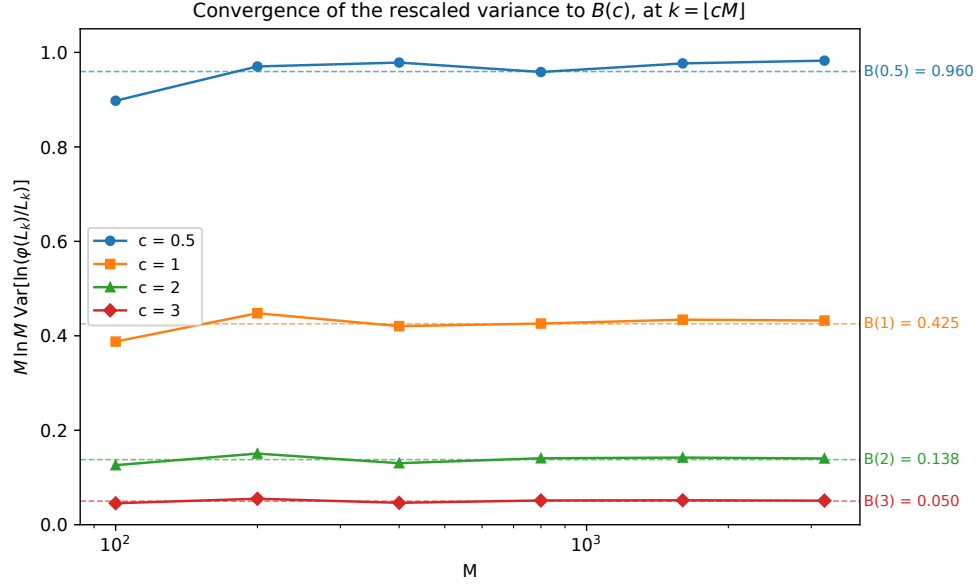


Figure 1: Rescaled variance  $M \ln M \cdot s_k^2$  versus  $M$  at  $k = \lfloor cM \rfloor$  for  $c \in \{0.5, 1, 2, 3\}$ . Values evaluated from the exact finite formula  $s_k^2 = V_1 + V_2$  using extended-precision arithmetic. Dashed horizontal lines indicate the limit  $B(c) = (e^c - 1)^{-1} - (e^{2c} - 1)^{-1}$  predicted by Theorem 1.

Table 1: Deterministic extended-precision evaluation of  $M \ln M \cdot s_k^2$  at  $k = \lfloor cM \rfloor$  across  $c \in \{0.5, 1, 2, 3\}$  and  $M \in \{100, 200, 400, 800, 1600, 3200\}$ , alongside the limiting value  $B(c)$  from Theorem 1. Values computed via  $s_k^2 = V_1 + V_2$  by direct summation; no Monte Carlo.

| $M$    | $c = 0.5$ | $c = 1.0$ | $c = 2.0$ | $c = 3.0$ |
|--------|-----------|-----------|-----------|-----------|
| 100    | 0.8977    | 0.3877    | 0.1261    | 0.0454    |
| 200    | 0.9702    | 0.4478    | 0.1506    | 0.0550    |
| 400    | 0.9785    | 0.4205    | 0.1303    | 0.0464    |
| 800    | 0.9585    | 0.4259    | 0.1407    | 0.0514    |
| 1600   | 0.9767    | 0.4342    | 0.1422    | 0.0518    |
| 3200   | 0.9826    | 0.4323    | 0.1403    | 0.0510    |
| $B(c)$ | 0.9595    | 0.4255    | 0.1379    | 0.0499    |

*Theorem 2.* Figure 2 displays the empirical distribution of the standardised coprime density  $(\varphi(L_k)/L_k - \hat{\rho}_k)/\hat{\sigma}_k$  at  $k = M$  for  $M \in \{100, 300, 1000\}$ , where  $\hat{\rho}_k, \hat{\sigma}_k$  are the sample mean and sample standard deviation. Sample sizes are  $N = 3000$  at  $M = 100$  and  $M = 300$ ;  $N = 1000$  at  $M = 1000$ . Sample skewness decreases from 0.61 at  $M = 100$  through 0.27 at  $M = 300$  to 0.16 at  $M = 1000$ ; excess kurtosis (Fisher convention) decreases from 0.58 at  $M = 100$  through 0.12 at  $M = 300$  to  $-0.18$  at  $M = 1000$ . These are descriptive diagnostics rather than statistical tests; they are consistent with the proof above.

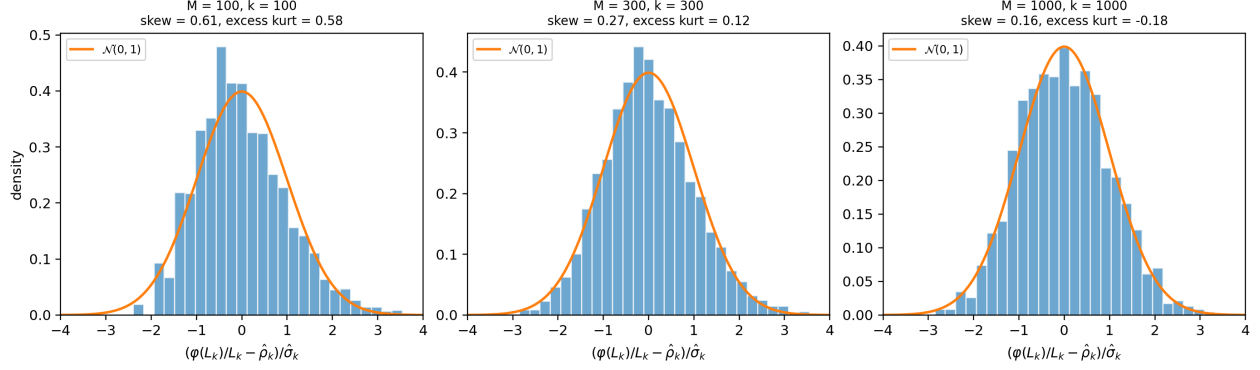


Figure 2: Empirical distribution of the standardised coprime density  $(\varphi(L_k)/L_k - \hat{\rho}_k)/\hat{\sigma}_k$  at  $k = M$ , for three values of  $M$ . Solid curve: standard normal density. Sample sizes  $N$  and Fisher-convention skewness/excess kurtosis reported in panel titles. Histograms use 35 bins in  $[-4, 4]$  and are normalised to density.

## 9 Discussion

*Why the increment bound needs both ingredients.* The trivial Mertens-type bound  $|D_i| \leq 2\Lambda(M) = O(\log \log M)$  is too large to feed into a Lyapunov condition:  $(\log \log M)^2 \cdot M \cdot \ln M$  grows, not shrinks. A bound using only the coupon-collector estimate (restricting to primes with  $m_p < (\log M)^2$ , ignoring the sieve) yields  $O(\log \log M)$  contributions per coordinate and is likewise insufficient. A bound using only the sieve-geometry (at most one contributing prime divides  $x$ ) without the coupon-collector filter allows primes with very small  $q_p$  (i.e. very large  $m_p$ ) that contribute as  $O(1/\pi_p) = O(\log \log M)$  to the worst-case  $|S_i(x)|$ . Both estimates together reduce  $|S_i(x)|$  to  $(\log M)^2/(M-1)$ , with the mean-correction  $|C_i|$  contributing an additional  $O((\log M)^3/M)$ . The resulting uniform bound  $\max_i |D_i| = O((\log M)^3/M)$  on  $\Omega^*$  is what the Hall–Heyde hypotheses need.

*Finite- $M$  convergence in Theorem 1.* The proof establishes convergence but does not optimise a uniform remainder for the diagonal prime-level sum. The numerical values in Table 1 show slow, mildly oscillatory convergence at the tested values of  $M$ . A quantitative rate would require uniform error estimates in the partial-summation argument of Lemma 6.

*General distributions.* The Doob decomposition (Theorem 11) generalises verbatim to any distribution on  $\{2, \dots, M\}$ , with  $\zeta_p^{(i)}$  redefined as  $\mathbf{1}\{p \mid X_i\} - \mathbb{P}(p \mid X)$ . The increment bound uses the uniformity of the  $X_i$  in two places: the coupon-collector estimate (which goes through if  $\mathbb{P}(p \mid X) \gtrsim m_p/M$  uniformly) and the sieve-geometric observation (which is distribution-independent once levels are defined by  $q_p$ ). The geometric disjointness in Proposition 5 is distribution-independent for any law supported on  $\{2, \dots, M\}$ : no support point is divisible by both  $p$  and  $q$  when  $pq > M$ . What is specific to the uniform law is the explicit formula  $\pi_p = m_p/(M-1)$  and the resulting level structure; a non-uniform extension would require distribution-specific estimates for these probabilities and covariances. Extension to product distributions close to uniform appears feasible; substantial departures from uniformity would change the limit  $B(c)$ .

*Comparison with fixed- $k$  CLT results.* Buraczewski–Iksanov–Marynych [3] established a CLT for  $\log \text{lcm}$  of a uniform  $m$ -tuple on  $\{1, \dots, n\}$  with  $m$  fixed as  $n \rightarrow \infty$ . Our result is not comparable in regime: they study  $\log \text{lcm}$  itself, with  $k$  fixed, while we study  $\log(\varphi(L_k)/L_k)$  with  $k = \lfloor cM \rfloor$  and  $M \rightarrow \infty$ . The natural common ground — a CLT that interpolates between the two regimes,

for instance  $k = f(M)M$  with  $f \rightarrow 0$  slowly — is not addressed here.

## 10 Conclusion

We have established, in the joint regime  $k = \lfloor cM \rfloor$ ,  $M \rightarrow \infty$ , fixed  $c > 0$ , the variance asymptotic  $M \ln M \cdot s_k^2 \rightarrow B(c) = (e^c - 1)^{-1} - (e^{2c} - 1)^{-1}$  (Theorem 1) and a central limit theorem for the standardised log coprime density (Theorem 2). The variance theorem is proved by a direct diagonal / off-diagonal decomposition, with the off-diagonal term controlled by an exact covariance identity (Lemma 7) valid for prime pairs  $(p, q)$  with  $pq > M$ . The CLT is proved via the Hall–Heyde martingale central limit theorem applied to the Doob filtration. Its two nontrivial ingredients are a uniform increment bound combining coupon-collector concentration with a sieve-geometric observation (Theorem 12), and a conditional-variance concentration resting on per-prime moment bounds, the covariance estimate of Lemma 15, and an  $L^2$  Minkowski bound that controls all prime-pair cross terms. Together with the first-order results of the companion paper [12], these give a two-moment asymptotic theory for the coprime density in this regime.

**Data and code availability.** The numerical code, fixed random seed, generated figures, and machine-readable outputs underlying Table 1 and Figures 1–2 are openly available at

<https://github.com/Wakid90/Variance-and-Central-Limit-Theorem-for-the-Log-Coprime-Density-of-Random-LCMs>

The repository includes a one-command reproduction script and an arbitrary-precision validation mode for the exact variance formula.

## References

- [1] G. Alsmeyer, Z. Kabluchko, A. Marynych, *Limit theorems for the least common multiple of a random set of integers*, Trans. Amer. Math. Soc. 372 (2019), 4585–4603.
- [2] A. Bostan, A. Marynych, K. Raschel, *On the least common multiple of several random integers*, J. Number Theory 204 (2019), 113–133.
- [3] D. Buraczewski, A. Iksanov, A. Marynych, *Central limit theorem for the least common multiple of a uniformly sampled  $m$ -tuple of integers*, J. Number Theory 233 (2022), 301–336.
- [4] J. Cilleruelo, J. Rué, P. Šarka, A. Zumalacárregui, *The least common multiple of random sets of positive integers*, J. Number Theory 144 (2014), 92–104.
- [5] P. Hall, C. C. Heyde, *Martingale Limit Theory and Its Application*, Academic Press, New York, 1980.
- [6] T. Hilberdink, L. Tóth, *On the average value of the least common multiple of  $k$  positive integers*, J. Number Theory 169 (2016), 327–341.
- [7] A. Iksanov, A. Marynych, K. Raschel, *Asymptotics of arithmetic functions of GCD and LCM of random integers in hyperbolic regions*, Results Math. 77 (2022), article 165.
- [8] Z. Kabluchko, A. Marynych, K. Raschel, *Multivariate multiplicative functions of uniform random vectors in large integer domains*, Results Math. 78 (2023), article 201.

- [9] J. B. Rosser, L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. 6 (1962), 64–94.
- [10] C. Sanna, *On the least common multiple of random  $q$ -integers*, Res. Number Theory 7 (2021), article 16.
- [11] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Graduate Studies in Mathematics 163, Amer. Math. Soc., 2015.
- [12] M. Wakid, *On the Expected Coprime Density of the Least Common Multiple of Random Integers*, Zenodo (2026), [doi:10.5281/zenodo.21325880](https://doi.org/10.5281/zenodo.21325880).